

MASTER DIGITAL FORENSICS

Complete Textbook, University Course & Practitioner Handbook

From Beginner to Expert Forensic Investigator

16 MODULES

20 ELEMENTS EACH

300+ TOOLS

EXPERT LEVEL

Law Enforcement	Military Intelligence	Incident Response
Cybercrime Investigation	eDiscovery	DFIR Consulting
Malware Analysis	Cloud Investigation	Expert Witness

*Covers: Digital Evidence • OS Forensics • File Systems • Data Acquisition •
Network • Email • Mobile • Cloud • Memory • Malware • Anti-Forensics •
Legal Frameworks • Incident Response • Emerging Technologies*

TABLE OF CONTENTS

MODULE 01	Introduction to Digital Forensics	History, principles, career paths, methodology
MODULE 02	Digital Evidence Fundamentals	Evidence types, chain of custody, integrity, admissibility
MODULE 03	OS Fundamentals for Digital Forensics	Windows/Linux/macOS internals, artifacts, registry
MODULE 04	Storage Media and File Systems	HDD/SSD/RAID, FAT/NTFS/EXT4, carving, slack space
MODULE 05	Data Acquisition and Preservation	Imaging tools, write blockers, hashing, legal hold
MODULE 06	Digital Artifact Analysis	Browser, email, USB, LNK, prefetch, event logs
MODULE 07	Network Forensics Fundamentals	Packet capture, protocol analysis, traffic reconstruction
MODULE 08	Email and Messaging Forensics	Headers, metadata, webmail, encrypted messaging
MODULE 09	Mobile Device Forensics	iOS/Android acquisition, chip-off, app analysis
MODULE 10	Cloud and Virtual Environment Forensics	AWS/Azure/GCP, VMs, containers, log analysis
MODULE 11	Memory and Malware Forensics	RAM acquisition, Volatility, malware analysis, IOCs
MODULE 12	Steganography and Anti-Forensic Techniques	Steganography, wiping, encryption, timestomping
MODULE 13	Legal Frameworks and Compliance	Laws, warrants, chain of custody, court testimony
MODULE 14	Incident Response and Cybersecurity Integration	IR lifecycle, SIEM, threat hunting, playbooks
MODULE 15	Future Trends and Emerging Technologies	AI forensics, IoT, blockchain, quantum, deepfakes
MODULE 16	Case Studies, Capstone and Certification Prep	Real cases, exam prep, career roadmap

01 Introduction to Digital Forensics

1. WHAT IS DIGITAL FORENSICS?

Digital Forensics (DF) is the scientific discipline of identifying, collecting, preserving, examining, analyzing, and presenting digital evidence in a manner that is legally defensible.

Formally defined by SWGDE: 'The application of science to the identification, collection, examination, and analysis of data while preserving the information and maintaining a strict chain of custody.'

2. CORE PRINCIPLES

Repeatability	Any qualified examiner should reach the same conclusion
Reproducibility	Results can be replicated on different systems
Integrity	Evidence must be unchanged from original (hashing)
Justifiability	Every forensic decision must be documented

02 Digital Evidence Fundamentals

1. WHAT IS DIGITAL EVIDENCE?

Digital evidence is any information stored or transmitted in digital form that a party to a legal case may use at trial.

2. TYPES OF DIGITAL EVIDENCE

Computer-stored records	Files, databases, emails, documents
Computer-generated records	Log files, system events, network flows
Volatile evidence	RAM contents, running processes (lost on power off)
Non-volatile evidence	Hard drives, SSDs, USB drives

3. CHAIN OF CUSTODY

Chain of Custody (CoC) is the documented, unbroken transfer of evidence from seizure to court presentation.

■ CHECKLIST: Evidence Handling

- Every item assigned unique exhibit number before handling
- Photograph each item in place before removal
- Document physical condition (damage, stickers, serial numbers)
- Place in appropriate packaging (anti-static for electronics)
- Complete chain of custody form at time of seizure
- Calculate MD5 AND SHA256 before analysis begins
- Record hash values in CoC and all forensic reports
- Secure evidence in locked storage with limited access

03 Module 3 Overview

Content for Module 3 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

04 Module 4 Overview

Content for Module 4 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

05 Module 5 Overview

Content for Module 5 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

06 Module 6 Overview

Content for Module 6 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

07 Module 7 Overview

Content for Module 7 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

08 Module 8 Overview

Content for Module 8 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

09 Module 9 Overview

Content for Module 9 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

10

Module 10 Overview

Content for Module 10 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

11

Module 11 Overview

Content for Module 11 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

12

Module 12 Overview

Content for Module 12 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

13

Module 13 Overview

Content for Module 13 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

14

Module 14 Overview

Content for Module 14 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

15

Module 15 Overview

Content for Module 15 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.

16 Module 16 Overview

Content for Module 16 covers advanced concepts, tools, and methodologies pertinent to its domain. Topics encompass best practices, common pitfalls, evidence collection strategies, and analysis tools.

KEY LEARNING POINTS

- Understand the architecture and footprint of the target environment.
- Identify and extract artifacts without altering original evidence.
- Utilize industry-standard tools for parsing, carving, and analysis.
- Document all findings meticulously to ensure legal admissibility.